# Dallin Jacobs

dallin62@byu.edu | 385.549.6006

## Education

**Brigham Young University — Provo, UT** Expected April 2027

*B.S. Machine Learning | GPA: 4.0*

Courses: Transformers for NLP, Machine Learning, Deep Learning, Reinforcement Learning

## Research Experience

**Record Linking Lab — Provo, UT** January 2026–Present

*Computer Vision Team Lead | Advisor: Dr. Joe Price*

- Led a five-person team collaborating with faculty from five universities on deliverables and technical trade-offs
- Extracted 7.9 million Paris census records from the 1930s for research and family-tree expansion
- Processed 222,000 land-status images and developed robust validation to identify and resolve silent-failures
- Improved WWI draft image extraction accuracy from 73% to 97% using an ensemble of 15 fine-tuned models
- Optimized LLM OCR efficiency using vLLM to gain a 112x throughput increase

**Perception, Control, and Cognition Lab — Provo, UT** February 2026–Present

*Research Assistant | Advisor: Dr. David Wingate*

- Built an ethical question extraction pipeline for 3.2 million ChatGPT conversations
- Explored and benchmarked limitations in 27 LLMs relating to ethics and faith
- Applied LLM interpretability analysis to explore religious omission in AI conversations

## Engineering Experience

**Red Compass Health — Draper, UT** December 2024–Present

*AI and Software Engineer*

- Designed and built AI-driven medical charting software to generate custom medical notes and treatment plans
- Integrated EHR and CRM APIs to automate data synchronization and appointment reminders

**Argus Home— Provo, UT** January 2026–Present

*Cofounder and Software Engineer*

- Founded a startup to design and manufacture an oven smart plug to reduce fire danger in homes

## Research Paper

**Coauthor:** Omissive Bias in Religious Representation: Benchmarking LLM Answers to Everyday Ethical Decision-making. arXiv:2605.24319 (2026) (preprint)

## Technical Skills

**ML:** PyTorch, Hugging Face, vLLM, fine-tuning, LLM quantization and optimization, LLM evaluation, OCR, agentic harnesses, context engineering

**Engineering:** Python, Slurm, Git, APIs, image segmentation, agentic coding

## Leadership and Service

**The Church of Jesus Christ of Latter-day Saints** August 2021–July 2023

*Volunteer Representative | Los Angeles, CA*

- Mentored groups of 8 missionaries; provided welfare service and taught principles of faith and responsibility

**Brigham Young University AI Club** September 2026–Present

*Leader | Provo, UT*

- Coordinated with faculty to teach students responsible and effective AI use
- Organized hackathons, presentations, and collaboration relating to AI concepts